



Basics of Civil Engineering

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Introduction to Civil Engineering

- Deals with engineered construction of buildings, roads, bridges, dams, tunnels, etc.
- What Civil Engineering Profession offers?
 - Enjoyable spacious and comfortable living in a building or underground enclave
 - Driving from a given location to virtually any region in the continent
 - Availability of plenty of Clean water for domestic use
 - And many more

Scope of Civil Engineering

- Planning
- Designing
- Estimating
- Supervising and
- Management
of different construction activities



Disciplines of Civil Engineering

1. Building Construction
2. Advanced Building Construction
3. Building Planning and Management
4. Environmental Engineering or Water Supply and Sanitary Engineering
5. Geotechnical Engineering (Soil Mechanics)
6. Surveying and Levelling
7. Structural Engineering
8. Transportation Engineering
9. Town Planning
10. Water Resource Engineering



1. Building Construction

- Construction of various structures and different types of buildings
- Construction of building components like brick work, Reinforced Cement Concrete (RCC) works, foundations, doors and windows, floor, roof, etc.
- Study of engineering materials like cement, steel, timber, glass, etc.



2. Advanced Building Construction

- Study of construction of deep foundations
- Underwater construction
- Construction of dams, bridges, tunnels, off-shore oil rig, etc.
- Study of equipment like power shovel, rollers, cranes etc. for large scale projects

3. Building Planning and Management

- Fundamental principles of planning
- Building bye-laws
- Planning of public, residential and industrial buildings
- Construction management of whole project
- Management of construction materials, labour and equipment

4. Environmental Engineering

- Design, construction and maintenance of water treatment plant, water distribution system and sewage system
- Waste water treatment and solid waste management
- Air, water and land pollution

5. Geotechnical Engineering

- Soil investigation
- Design of foundations
- Measurement of soil parameters and safe bearing capacity of soil
- Study of geology



6.Surveying and Levelling

- For setting out of works
- For preparing maps of land
- Measurement of levels of land and prepare contour maps
- Carried out using chains, compass, plane table, levels, the odolites, EDM instruments, etc.

7. Structural Engineering

- Design of RCC structures (Retaining walls, water tanks, bridges, residential buildings, etc.)
- Design of steel structures (railway platform, factory sheds, steel bridges, etc.)
- Design of earth quake resistant structures
- Concrete technology

8. Transportation Engineering

- Airport engineering
- Bridge engineering
- Harbour and Docks engineering
- Highway engineering
- Railway engineering
- Tunnel engineering
- Traffic engineering

9. Town Planning

- Arrangement of various components of a town in such a way that the town attains significance of a living organism
- Towns are divided into different zones like residential zone, commercial zone, industrial zone, etc.

10. Water Resources Engineering

- Fluid Mechanics-
deals with behaviour of all, more or less incompressible, liquids and gases
- Hydrology –
deals with study of sources of water, measurement and study of rainfall, flood, etc.
- Irrigation Engineering –
deals with designing of hydraulic structures like dams, canals, etc.

Relevance of Civil Engineering in Overall Infrastructural Development

- Buildings are designed by civil engineers by conveying an array of structural elements that support the architectural spatial distribution.
- Highway is designed by creating a plane that adequately supports weight and stresses of vehicles in motion.
- The water resource system is designed by a civil engineer that arranges for water intake at an appropriate source, a water conveyance set and a water distribution network.
- The civil engineer designs waste water disposal and waste water treatment plants.

Relevance of Civil Engineering in Overall Infrastructural Development

- Operations management for infrastructural development has three major aspects:
 - i. Quality
 - ii. Cost and
 - iii. Time

Relevance of Civil Engineering in Overall Infrastructural Development

- Infrastructure Development involves fundamental structures that are required for the functioning of a community & society
- This is usually referred to structures like roads, water supply, sewers, electrical grids, telecommunications, renewable energy, water sources identification & boring (wells), purification systems for clean water, hazard waste management and so on.
- Governments cannot manage to do it without skilled/trained manpower (which includes engineers from all the departments)

Relevance of Civil Engineering in Overall Infrastructural Development

- Civil engineers can manage to do these projects related to infrastructural development as the fundamental idea of less space and more efficiency is embedded in all civil engineers which is a crucial factor in development of infrastructure.
- So civil engineers have a very crucial role to play in the development of infrastructure

Relevance of Civil Engineering in Overall Infrastructural Development

The role of Civil Engineering activities in the infrastructural development can be summarised as follows:

- A proper planning of towns and extension areas in the cities.
- Fast rate of urbanisation and increase in the cost of land has forced civil engineers to go for vertical growth in cities
- This has resulted in new building technologies and sophisticated analysis methods.
- Civil engineers have to solve problems of rural areas as well.

Relevance of Civil Engineering in Overall Infrastructural Development

The role of Civil Engineering activities in the infrastructural development can be summarised as follows:

- Low cost housing is the need of the hour to make poor people afford their own houses.
- Water is an important need for all living beings.
- Civil engineers have to explore into various water resources and ensure water supply to urban areas throughout the year. Water is required for agriculture also.
- Good roadways and transportation facilities include another important amenity of the public which civil engineers deliver.
- Other important infrastructural activities of civil engineers are controlling pollution of air, water and land.

Relevance of Civil Engineering in Overall Infrastructural Development

The impact of infrastructural development of a country

- Provide protection from drought, famine, flood, etc.
- Improved irrigation facilities
- Better sewage system
- Improved education facilities
- Improvement in transportation and communication
- Generation of electricity from natural resources



INTRODUCTION TO TYPES OF BUILDING AS PER NBC



Types of Buildings as per National Building Code (NBC)

- ✓ NBC is prepared to unify the building regulations throughout the country.
- ✓ NBC is prepared by Bureau of Indian Standards (BIS).
- ✓ First version is published in 1970.
- ✓ This is revised in 1983.
- ✓ The current edition is updated in 2016.

Types of Buildings as per National Building Code (NBC)

- Nine groups based on their Occupancy
 - Group A - Residential buildings
 - Group B - Educational buildings
 - Group C - Institutional buildings
 - Group D - Assembly buildings
 - Group E - Business buildings
 - Group F - Mercantile buildings
 - Group G - Industrial buildings
 - Group H - Storage buildings
 - Group J - Hazardous buildings

Group A: Residential Buildings



Buildings in which sleeping accommodation provided for normal residential purposes with or without cooking or dining or both facilities, except any building classified under Group C.



Group B: Educational Buildings



- Buildings used for school, college, other training institutions for day-care purposes involving assembly for instruction, education or recreation for not less than 20 students.
- Residential accommodation provided in school/ institution are classified as sub-division A-3.



Group C: Institutional Buildings



➤ Any building or part thereof, which is used for purposes, such as **medical or other treatment or care of persons** suffering from physical or mental illness, disease or infirmity; cares of infants, convalescents or aged persons and for penal or correctional detention in which the liberty of inmates is restricted.



Group D: Assembly Buildings



- Any building or part of building, where number of persons **not less than 50** **congregate or gather** for amusement, recreation, social, religious, patriotic, civil, travel and similar purposes.
- Eg: theatres, assembly hall, museum, gymnasiums, restaurants, places for worships etc



Group E: Business Buildings



- Any building or part of a building, used for **transaction of business** (other than that covered by building in group F, **for keeping accounts and records** and similar purposes.
- Barber shops, beauty parlours are included in this group.



Group F: Mercantile Buildings



➤ Any building or part of a building, used as shops, stores, market, for display and sale of merchandise, either wholesale or retail



Group G: Industrial Buildings



- Any building or part of a building or structure, in which products or materials of all kinds and properties are fabricated, assembled, manufactured or processed.
- Assembly plants, industrial laboratories, dry cleaning plants, power plants generating units, pumping stations, fumigation chambers, laundries, building or structures in gas plants, etc.



Group H: Storage Buildings



➤ Any building or a part of a building used primarily for or the **storage or sheltering** (including servicing, processing, or repairing incidental to storage) **of goods, wares or merchandise** (except that those involve highly combustible or explosive products or materials) **vehicles or animals**.

➤ Warehouses, cold storage, grain elevators, freight depots, barns, transit sheds, storehouse, garage, hangers etc



Group J: Hazardous Buildings



- Used for the storage, handling, manufacture or processing of highly combustible explosive materials
- Which are liable to burn with extreme rapidity and which may produce poisonous fumes or explosions for storage, handling, manufacturing or processing which involve highly corrosive, toxic alkalis, acids or other liquids or chemical producing flames, fumes and explosive, poisonous, irritant or corrosive gases.



Occupancy classification of buildings



Group	Occupancy	Sub divisions
A	Residential	A1 Lodging or rooming houses A2 One or two family dwellings A3 Dormitories A4 Apartments A5 Hotels
B	Educational	Building used for colleges, schools, day care purpose involving assembly for instructions, education etc



Occupancy classification of buildings



Group	Occupancy	Sub divisions
C	Institutional	C1 Hospitals and sanatoria C2 Homes for aged, orphans C3 Prisons, mental hospitals
D	Assembly	D1 Theatres with fixed seats for more than 1000 persons D2 Theatres with fixed seats for less than 1000 persons D3 Halls with capacity to hold more than 300 persons but with no permanent seating



Occupancy classification of buildings



Group	Occupancy	Sub divisions
D	Assembly	D4 Halls for less than 300 persons but with no permanent seating D5 All other assembly buildings not covered above
E	Business	E1 Office, banks, professional establishments for doctors, engineers, lawyers E2 Laboratories and research E3 Computer installations



Occupancy classification of buildings



Group	Occupancy	Sub divisions
F	Mercantile	F1 Shops, stores, markets with area up to 500m ² with storage incidental to sales F2 Underground shopping centres/ departmental stores with area more than 500m ² with storage and services incidental to sales
G	Industrial	G1 Buildings used for low hazard industries G2 Buildings used for moderate hazard industries G3 Buildings used for high hazard industries



Occupancy classification of buildings



Group	Occupancy	Sub divisions
H	Storage buildings	Used for storage or sheltering of goods, vehicles, animals Warehouses, grain stores, garages, stables
J	Hazardous	Used for storage, handling, manufacturing and processing of corrosive, toxic, inflammable, explosive materials



Selection of Site for Buildings

The following general principles should be kept in mind while selecting a site for a building:

- The **purpose** of building and extend of **privacy** is desired
- The site should be located in **fully developed or fast developing locality**
- The site should be located in such a way that **community services** like police and fire protection, clearing of waste and street cleaning, **utility services** like water-supply, electricity, etc., **amenities** like school, hospital, market, etc. and means of **transport** are also available
- Before selecting a site, one should study the **bye-laws** of the local authority, which put before restrictions regarding proportions of plots to be built-up, open spaces and margins to be left around, heights, etc.
- Area of plot should be such that the building constructed on it meets the requirement of the owner, after following certain restrictions of local authority

Selection of Site for Buildings

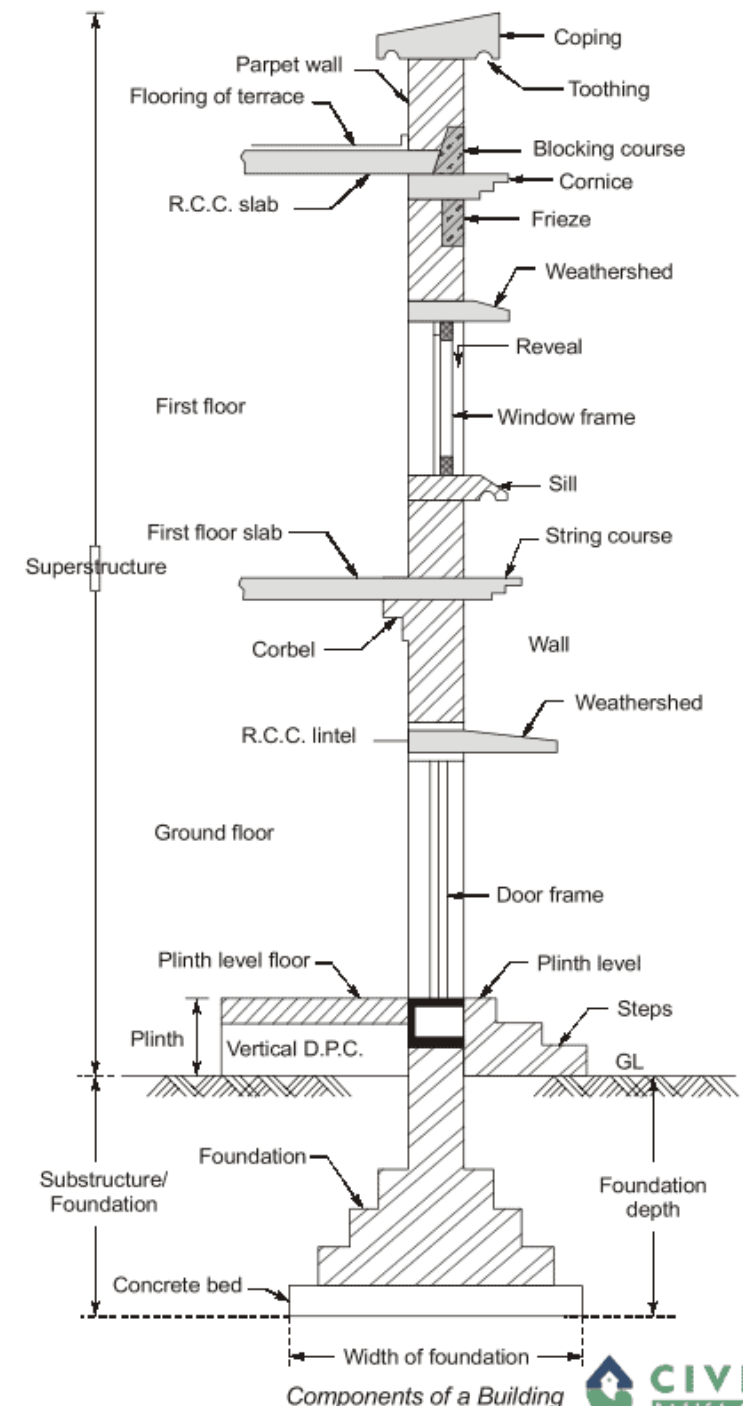
The following general principles should be kept in mind while selecting a site for a building:

- **Shape of the plot should not be irregular**, and should not be having any sharp corners
- The site should be **situated on an elevated place** and also levelled with uniform slopes from one end to the other to **provide good drainage of rain water**
- The **soil** of the site should be good enough with **high Safe Bearing Capacity** to provide economical foundations
- The site should be situated in a naturally beautiful environment which creates **healthy living and working conditions**
- The site should be **away from quarries, kilns, factories, industries, rivers with heavy flood, etc.**, if possible
- The **legal and financial aspects** should also be given due consideration

Components of a residential building

Major Parts of a building

- **Foundation** - part of a building constructed below ground level and which is in direct contact with sub-strata and transmits all the loads to the sub-soil
- **Plinth** - building above the ground level and up to the floor level immediately above the ground
- **Superstructure** - part of the building constructed above the plinth level.



Building Component	Functions
Foundation	Transmits the loads; supports the superstructure; provides stability; Provides Safety against scouring
Plinth	Helps in transmitting loads from superstructure to substructure; Protects the building from moisture rainwater, dust, insects, termite, etc.
Wall	Supports beam and slab; transmits the loads. Provides partition, privacy and safety; Protects building against heat, cold, rain, noise, fire, etc.
Column	Supports beam and slab Transmit the loads
Floor	Gives a plane and levelled surface for the occupants
Roof	Covers the top of the building Gives protection against rain, heat, snow, sound, wind, etc

Building Component	Functions
Door	Permits entry, exit, light and ventilation to the building Imparts safety and privacy to the building
Window	Gives nice scenic view to the building Permits light and ventilation
Step	For access in building from GL to upper floors
Stair	For vertical circulation among the floors in the building
Lintel, Arch	Gives support to the wall above the openings in wall Distributes load evenly
Sill	Gives support to the bottom of the window openings
Beam	Supports the slab Transmits loads coming from slab to column or wall
Weather shed (Sun shade)	Protects the walls of the buildings from sun, heat and rain
Parapet	Provides boundary to the terrace and encloses it

Basic requirements of a building

- The planning and construction of a building should be aimed at fulfilling the following requirements
 1. Strength and stability
 2. Dimensional stability
 3. Resistance to dampness
 4. Resistance to fire
 5. Heat insulation
 6. Sound insulation
 7. Protection against termite attack
 8. Durability
 9. Security against burglary
 10. Lighting and ventilation
 11. Comforts and convenience
 12. Economy



PLANNING

- All buildings should be properly planned, keeping in view the various requirements of a good building.
- Except strength requirement, all other requirements of a good buildings are taken care at the stage of planning.
- Strength requirement is taken care during structural design of building components.
- However in planning the building **by-laws of the statutory authorities should not be violated**. Planning of the building is an art combined with science.

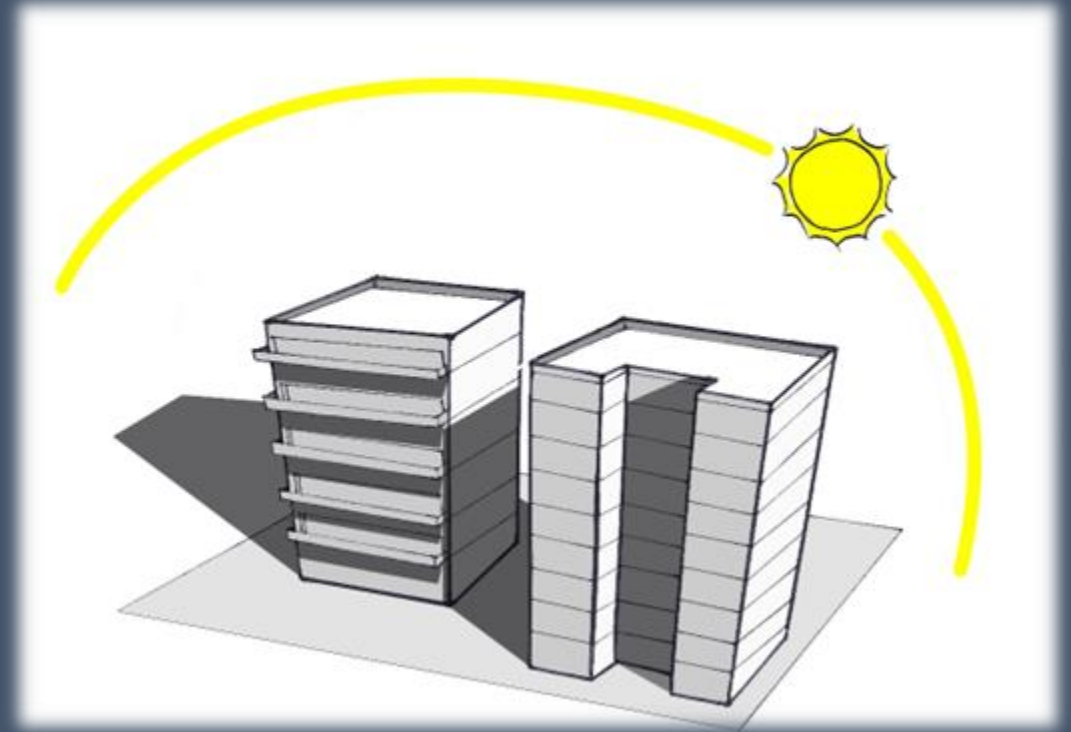
PLANNING

- Principles of planning of buildings
- 1. Orientation
- 2. Energy efficiency
- 3. Utility
- 4. Other requirements of the building



1.Planning suitable orientation

- Orientation means setting out the plan of the building with respect to north-south and east-west directions to provide an opportunity for user to enjoy sun-shine and breeze when required and to avoid the same whenever not required.
- This is also known as planning the **aspect of a building**.
- Aspect means arrangement of doors, windows in the external wall to make good use of nature.
- Kitchen should have eastern aspect to enjoy morning sunshine, means, kitchen should be located on the eastern side of the building to make use of morning sun rays.



1.Planning suitable orientation

- The following are the required aspects for various parts of the building in the northern hemisphere of earth
1. Kitchen—eastern aspect.
 2. Dining room—southern aspect to enjoy winter sun.
 3. Drawing and living room—southern or south-eastern aspect to enjoy winter sun.
 4. Bed rooms—western or south-western breeze in summer.
 5. Reading room, class room, stairs - northern aspect to enjoy diffused light



2. Planning for energy efficiency

- A building should be planned in such a manner that it gives **maximum day lighting, ventilation and heat insulation**.
- If these requirements are fulfilled, requirement of electric energy comes down.

Light:

Natural light provides hygienic atmosphere.

Light should not be glaring but it should be **uniformly distributed**.

Providing windows and ventilators of appropriate size at suitable positions contributes a lot for natural lighting.

For **residential buildings window area to floor area** should not be less than **1/10th** while for **school buildings** it should not be less than **1/5th of floor area**.

For factory buildings north light trusses should be provided to get maximum diffused light.



2. Planning for energy efficiency

- **Ventilation:**
- Ventilation is the **circulation of the air** in the building.
- Natural ventilation can be achieved by selecting and positioning of doors, windows and ventilators at suitable places.
- Always **cross ventilations** should be planned suitably.
- Provision of **ventilators at roof level helps in driving out hot airs**.
- In case it is not possible to achieve natural ventilation for any part of the building provide ordinary or exhaust fans.



2. Planning for energy efficiency

- Heat Insulation:
- Thicker exterior walls provide insulation against heat.
- Proper ventilation also helps in achieving heat insulation.
- Sun shades provided to doors, windows and ventilators help in achieving heat insulation.
- In factories and assembly halls, height should be more to reduce temperature inside the building.
- The position of furnaces in the factories should be located away from the other parts of the factory.
- The openings should be provided at higher level in the wall to remove hot air.

3. Planning for suitable utility

- Principles of planning for suitable utility are:
- 1. Roominess
- 2. Furniture Requirements
- 3. Groupings
- 4. Circulation

4. Planning for meeting other requirements

- Principle of planning involves planning for meeting the following requirements also:
- 1. Sanitary convenience
- 2. Prospects
- 3. Elegance
- 4. Flexibility
- 5. Privacy
- 6. Resistance to fire
- 7. Sound insulation
- 8. Protection from termite
- 9. Security against burglary
- 10. Economy
- 11. Provisions for future alterations.

Key Terminologies in Building Construction

In civil engineering, precise terminology is used to describe different parts of buildings and their measurements. Understanding these terms is crucial for both planning and construction:

➤ **Plinth Area**

- The plinth area is the total covered built-up area measured at the floor level of a building.
- It is the area occupied by the building on the ground, including walls and columns, but excluding open spaces like balconies and terraces.
- The plinth area is essential for calculating the building's total area and for determining costs.

➤ **Carpet Area**

- Carpet area refers to the usable floor area of a building, excluding the thickness of the inner walls and common areas like lobbies and corridors.
- It is the actual area where people can live or work within the building. Carpet area is usually smaller than the plinth area.

➤ **Floor Area**

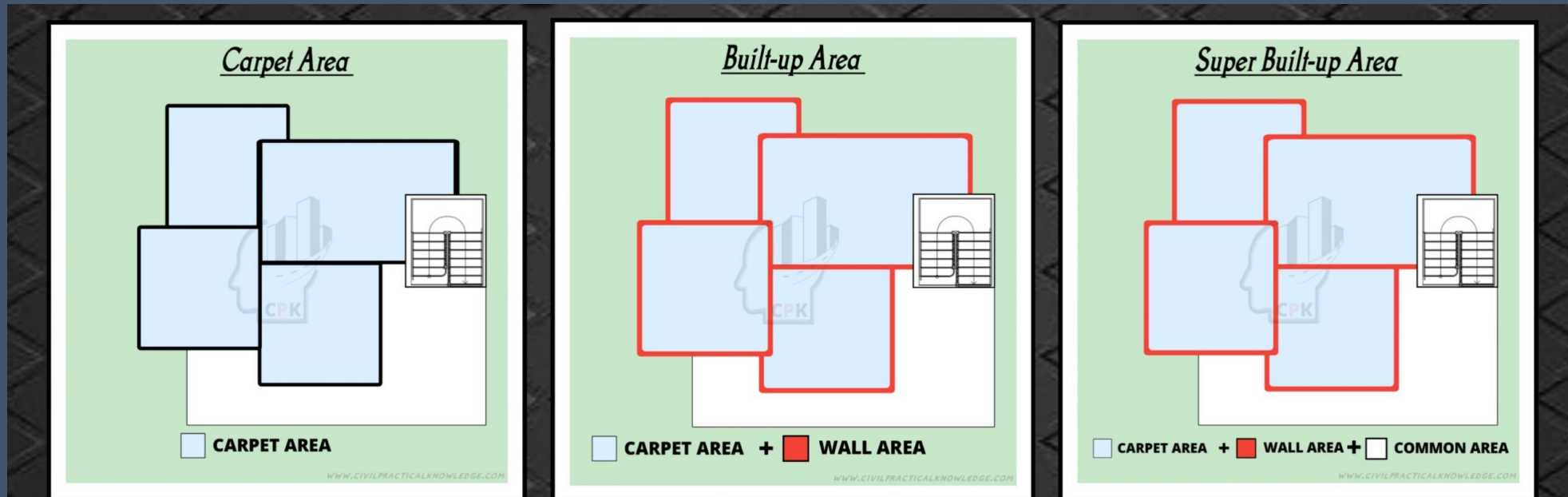
- The floor area is the total area of all floors in a building, measured from the external walls.
- This includes the usable area on all levels, from the ground floor to the top floor.
- Floor area is a critical metric for determining the building's capacity and design efficiency.

➤ Built-Up Area

- Built-up area is the total space covered by the building, including the carpet area, the thickness of external and internal walls, and utility spaces like balconies and verandas.
- It is larger than the carpet area and is a key consideration in property valuation and cost estimation.

➤ Floor Space Index (FSI)

- Floor Space Index (FSI), also known as Floor Area Ratio (FAR), is the ratio of a building's total built-up area to the total area of the plot on which it is built.
- FSI is used to regulate the height and density of buildings in urban planning.
- A higher FSI allows more construction on a given plot, which is typical in high-density urban areas.



THANK YOU !!!

Any Queries